

Stock Market Prediction Using Machine Learning

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G-0052

Abstract

Stock market prediction plays a crucial role in enabling investors to make informed decisions and maximize returns in a highly dynamic and volatile environment. Once reserved for a select few within the wealthiest of countries, stock market prediction is now more accessible than ever. This research aimed to determine if machine learning techniques could be used to predict future stock prices and direction by leveraging historical data and utilizing basic and hybrid models. NVIDIA stock data from 1999 to 2024 was used for experimentation. By implementing basic models such as RNN, CNN, LSTM, and GRU for pattern recognition, the goal was to forecast short-term price fluctuations and price trajectory to optimize investment strategies. Additionally, a combination of these models was used in the hopes of combining their best features into a more accurate prediction tool. Results indicated that all models had their strengths and weaknesses, but while the basic models performed better on average, the hybrid models showed great potential.

Methods

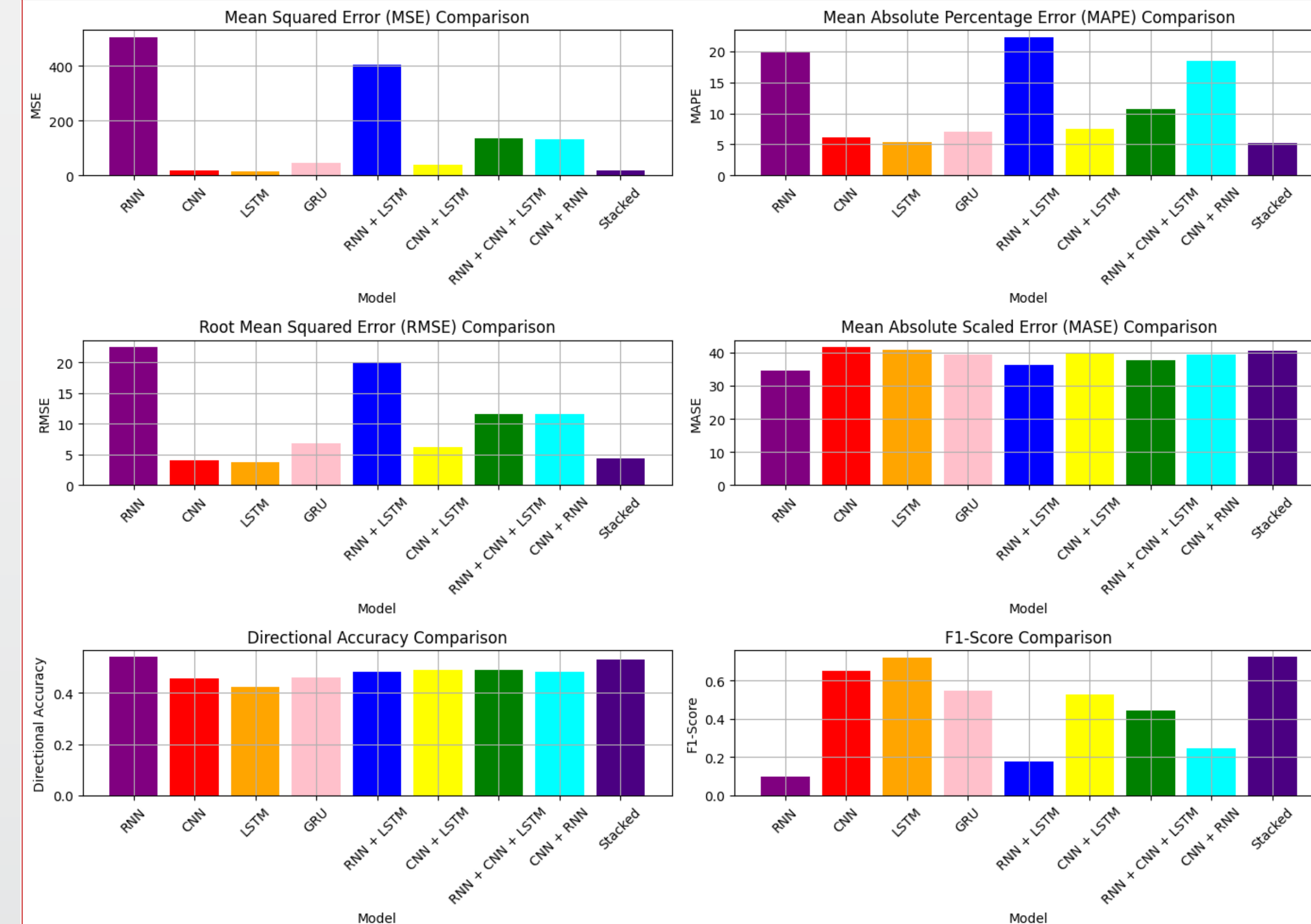
Data Collection and Processing:

- Historical stock price data gathered through Yahoo Finance API
- Data preprocessing: removal of irrelevant columns, handling missing values
- Rolling Window approach, models trained and tested on consecutive periods
- Feature engineering focusing on 6-month historical data with weighted importance

Model Architectures Used:

- Simple RNN Predictions
- CNN Predictions
- LSTM Predictions
- GRU Predictions
- CNN + LSTM Predictions
- CNN + GRU Predictions
- CNN + GRU + LSTM Predictions
- COMBINED Predictions
- Stacked Model Predictions

Results

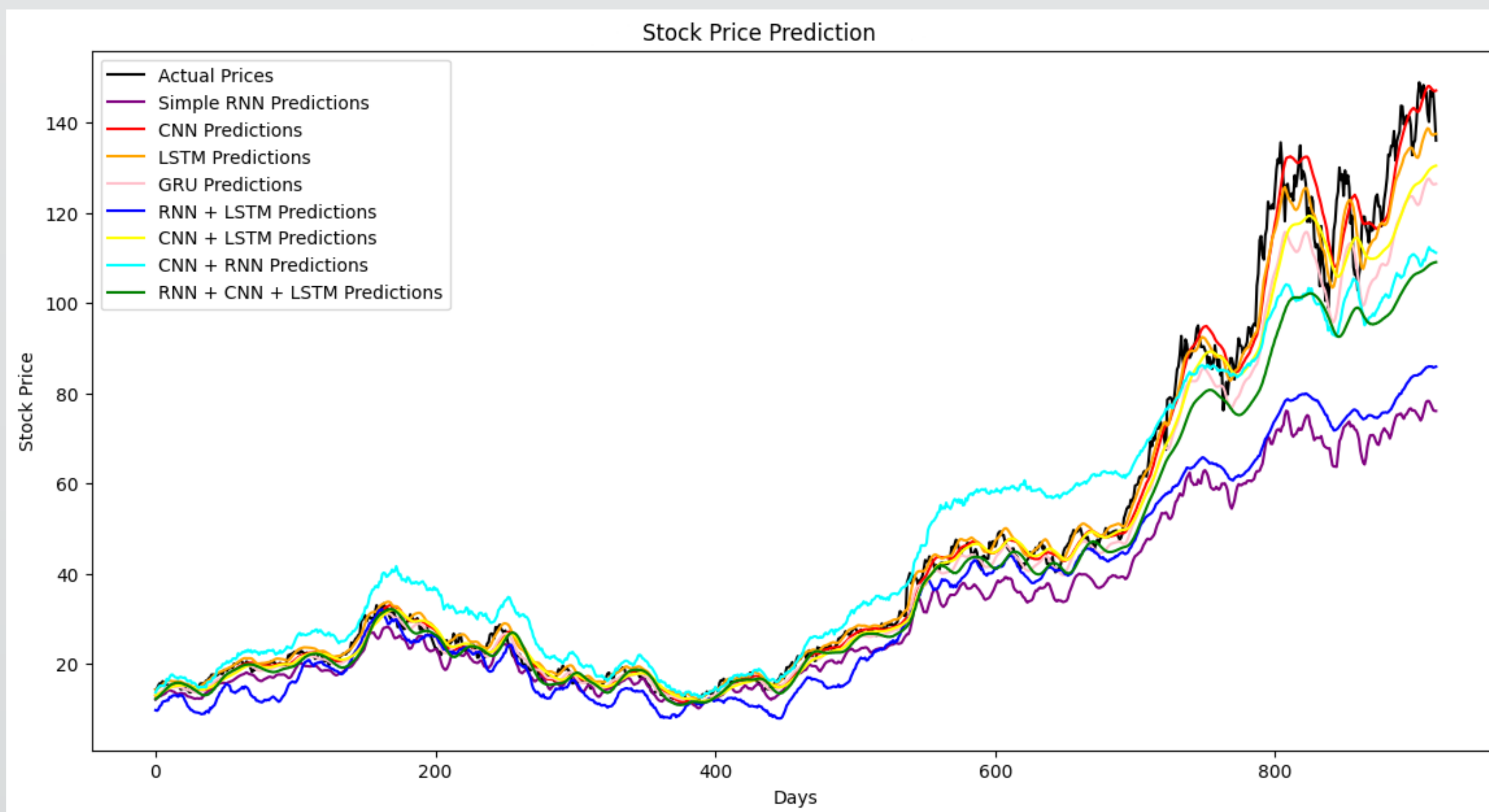


Background

- In stock market research, deep learning techniques are being increasingly applied to address various challenges.
- Traditional prediction methods often require specialized resources and information.
- Deep learning techniques have shown promise in financial forecasting
- Recent studies by Fischer & Krauss (2018) demonstrate LSTM effectiveness in processing sequential financial data.
- Complex and hybrid models show promise for improved performance.
- Combined CNN-LSTM approaches have proven successful in feature extraction and trend analysis.
- NVIDIA stock data from 1999 to 2024 will be used for experimentation.

Date	Open	High	Low	Close	Adj Close	Volume
1999-01-25	0.044271	0.045833	0.041016	0.045313	0.041559	510480000
1999-01-26	0.045833	0.046745	0.041146	0.041797	0.038334	343200000
1999-01-27	0.041927	0.042969	0.039583	0.041667	0.038215	244368000
1999-01-28	0.041667	0.041927	0.041276	0.041536	0.038095	227520000
1999-01-29	0.041536	0.041667	0.039583	0.039583	0.036303	244032000
...	...	...	...	...	...	...
2024-11-12	146.779999	149.649994	146.009995	148.289993	148.289993	198634700
2024-11-13	149.070007	149.330002	145.899994	146.270004	146.270004	191903300
2024-11-14	147.639999	149.000000	145.550003	146.759995	146.759995	194463300
2024-11-15	144.869995	145.240005	140.080002	141.979996	141.979996	249120000
2024-11-18	139.539993	141.550003	137.149994	140.149994	140.149994	216059761

6498 rows x 6 columns



Date	CNN	RNN	LSTM	GRU	CNN + LSTM	CNN + GRU	CNN + GRU + LSTM	Combined	Stacked	Actual Prices
2024-11-08	142.650848	74.567574	132.365234	121.927780	126.773178	119.587631	123.549606	133.343994	136.643738	147.630005
2024-11-11	143.532822	73.946671	133.471176	122.763977	127.054352	120.247002	123.673225	133.698303	137.925812	145.259995
2024-11-12	144.676559	74.499107	134.793930	123.952621	127.351227	120.875755	124.208862	134.576035	139.321548	148.289993
2024-11-13	145.533646	75.672409	135.876038	125.092140	127.768555	121.725403	124.496811	135.672241	140.356522	146.270004
2024-11-14	146.637268	76.925041	136.870163	126.080589	128.198746	122.448982	125.236572	136.588776	141.356750	146.759995
2024-11-15	147.400970	78.044594	137.686813	126.867928	128.671875	123.215492	125.641266	137.535583	142.192169	141.979996
2024-11-18	147.977890	78.273361	138.283493	127.433250	129.122452	123.810371	126.397591	138.280579	142.771042	140.149994
2024-11-19	148.087677	78.241714	138.243500	127.550476	129.558243	124.250572	126.793144	138.845413	142.617371	147.009995
2024-11-20	147.818329	77.716286	137.566803	127.107185	129.866302	124.339668	127.354691	138.944107	141.771713	145.889999
2024-11-21	147.388748	77.017319	137.114105	126.584602	130.128235	124.235275	127.629478	138.575638	141.316833	146.669998
2024-11-22	147.129303	76.240929	136.935944	126.270050	130.288956	124.166191	127.953506	138.227341	141.173523	141.949997
2024-11-25	146.949600	76.065811	137.051132	126.291840	130.449661	124.211693	128.133774	138.055634	141.283478	136.020004
2024-11-26	147.084824	76.049530	137.055542	126.390427	130.534561	124.092384	128.292053	138.068390	141.227417	136.919998
2024-11-27	146.649887	76.828094	135.548462	125.489288	130.573517	123.755974	128.482910	138.139389	141.227417	NaN

Conclusion

- Improved performance with early stopping and learning rate reduction
- Significant improvement by rearranging certain models such as switching from RNN + CNN + LSTM to CNN + RNN + LSTM
- Difficulty in finding models with minimized MSE without sacrificing directional accuracy.
- Hybrid models show promise, stack/combined models are superior

Future Direction

1. Feature Engineering: generate new features from existing data, such as technical indicators (e.g., moving averages, RSI, MACD), which can help the model capture more relevant patterns and make better predictions.
2. Train the model on more general stock trends rather than just one stock (NVDA)
3. Implement sentimental analysis
4. Give more weight to previous day volatility

Acknowledgments

We would like to thank our Data Science Professor, Dr. Nouhad Rizk, and her teacher's assistant, Jusvin Charles for their assistance and notes throughout the project.